

EXPERIMENT NO. 14:

Implement House Price Prediction using appropriate machine learning algorithm

Aim:

To implement a House Price Prediction Model in Python using Linear Regression and predict house prices based on suitable features.

Procedure:

1. Import the required Python libraries.
2. Create a sample house price dataset.
3. Separate the input features and target house price.
4. Split the dataset into training and testing sets.
5. Create and train the Linear Regression model.
6. Predict house prices using the test data.
7. Calculate the R^2 score to evaluate the model.
8. Predict the price of a new house.

Program:

House Price Prediction using Linear Regression

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Sample House Dataset
data = {
    'Area': [1000, 1200, 1500, 1800, 2000, 2200, 2500, 2800, 3000, 1600],
    'Bedrooms': [2, 2, 3, 3, 4, 4, 4, 5, 5, 3],
    'Bathrooms': [1, 2, 2, 2, 3, 3, 3, 4, 4, 2],
    'Age': [20, 15, 10, 8, 5, 5, 3, 2, 1, 12],
    'Price': [2500000, 3000000, 3800000, 4500000, 5200000,
              5800000, 6500000, 7200000, 8000000, 4000000]
}
```

```
df = pd.DataFrame(data)
```

```
# Input features and target
```

```
X = df[['Area', 'Bedrooms', 'Bathrooms', 'Age']]
```

```
y = df['Price']
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Create Linear Regression model
```

```
model = LinearRegression()
```

```
# Train the model
```

```
model.fit(X_train, y_train)
```

```
# Predict test data
```

```
y_pred = model.predict(X_test)
```

```
# Calculate R2 Score
```

```
score = r2_score(y_test, y_pred)
```

```
print("Actual Prices:")
```

```
print(y_test.values)
```

```
print("\nPredicted Prices:")
```

```
print(y_pred.round(2))
```

```
print("\nR2 Score:")
```

```
print(round(score, 2))
```

```
# Predict price of a new house
```

```
new_house = [[2000, 3, 2, 5]]
```

```
prediction = model.predict(new_house)
```

```
print("\nPredicted Price of New House:")
```

```
print(round(prediction[0], 2))
```

Output:

```
*** Actual Prices:
[8000000 3000000]

Predicted Prices:
[7781604.28 3012508.72]

R2 Score:
1.0

Predicted Price of New House:
5095466.17
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(
```

Result:

The House Price Prediction Model was successfully implemented in Python using Linear Regression. The model predicted house prices based on area, number of bedrooms, bathrooms, and age of the house, and successfully estimated the price of a new house.